

ARIEL ZINGER

Full-Stack Developer | Project Manager | Technical Product Leadership

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SUMMARY

After a decade leading R&D projects and cross-functional teams, I've recently completed an intensive full-stack and AI development bootcamp. I've gained hands-on skills in **Python, JavaScript/TypeScript, React, Node.js/Express, PostgreSQL, and agentic AI tools such as LangChain, LangGraph, and API/MCPs**, along with real experience directing AI-assisted development tools against clear specifications. AI is a genuine passion of mine; I'm drawn to the rapid pace the field moves and how much potential and value these tools unlock. That curiosity is backed by over a decade of R&D and technical leadership experience, having a strong foundation in project management, interdisciplinary collaboration, and turning complex problems into structured, deliverable outcomes.

TECHNICAL SKILLS

- **Languages:** Python, JavaScript, TypeScript, SQL
- **Frontend:** React, Redux Toolkit, HTML5, CSS3, Tailwind CSS
- **Backend:** Node.js, Express.js, RESTful API design, JWT authentication
- **Databases:** PostgreSQL, Knex.js query builder, relational schema design
- **AI & Agents:** Prompt engineering, LangChain, LangGraph, MCP, ReAct / TAO agent design patterns
- **Tools & Practices:** Git/GitHub, Agile/Scrum fundamentals, deployment basics (Render), AI-assisted development (Base44, Claude Code, Codex) and prompt-driven spec-to-code workflows, LLM-integrated features, Agentic workflows.

TECHNICAL MANAGEMENT SKILLS

- **Technical Specification:** database schema design, REST API design, system architecture documentation
- Product Lifecycle Management, **Go-to-Market (GTM)** Strategy, Market Segmentation, Competitive Intelligence
- **PRD/MRD** Authoring, Requirements Traceability, Roadmap Prioritization
- **Team Leadership & Cross-Functional Coordination**
- **Project Scoping**, Planning & Milestone Delivery
- **Budget & Resource Allocation**, Vendor/Supplier Management
- **Agile & Scrum** Fundamentals, Sprint Structure
- **Risk Management**, Process Documentation

EDUCATION

- **Full Sack & AI Bootcamp course** | Developers Institute | 2026
- **Product Management Certification** | Technion | 2025
- **Executive Program for Managers in DeepTech Startups** | Tel Aviv University & Israel Innovation Authority | 2025–2026
- **M.Sc. Chemistry (Materials & Interfaces)** | Weizmann Institute of Science, 2015
- **B.Sc. Chemistry (Extended Track)** | Tel Aviv University, 2012 — Top 10%

PROFESSIONAL BACKGROUND

Lead Scientist | Daika Ltd (2020 – 2024)

- Directed the technical product strategy and roadmap for the company's core wood-composite binder technology, overseeing the transition from proof-of-concept to industrial-scale production.
- Headed a multidisciplinary team of 10+ engineers, material scientists, and industrial designers.
- Collaborated directly with Bizdev and Marketing to align technical milestones with commercial growth targets.
- Oversaw equipment procurement, vendor relationships, and pilot-scale manufacturing setup.

R&D Senior Researcher & Project Manager | Nobio Ltd / Luminera Ltd (2015 – 2020)

- Coordinated interdisciplinary technical requirements gathering for dental- and dermal-material product development.
- Developed QC protocols, SOPs, and regulatory documentation to meet medical-device standards.
- Managed project timelines and interdepartmental coordination between R&D, regulatory, and production teams.

PERSONAL PROJECTS

CirqHub — AI-Driven Circular Economy Platform for Physical Markets — Hackathon project, 2026 — AI-assisted build

Co-designed the system architecture and led QA and code review for an AI-driven surplus-management platform for physical market vendors, built with Base44 as an AI-assisted development platform.

- Defined the system's core design: the item lifecycle and REST API surface, food-safety check constraints, and the platform's two AI decision engines — a real-time engine ranking surplus disposition (Donate / Barter / Reuse) and feeding recipe generation, and a predictive engine forecasting capacity to drive staffing and layout planning.
- Guided AI-assisted implementation of the PostgreSQL schema, Express API, and an offline-first React front end (custom offline-queue hook with local caching and background re-sync), working iteratively rather than accepting first-pass output.
- Performed manual QA across the vendor and operator surfaces — verifying disposition-state transitions, safety-constraint enforcement, and offline/online sync behavior — and code-reviewed the generated implementation for correctness and adherence to spec before sign-off.

<https://cirqhub.base44.app>

Biz2code — A Linear Gatekeeper for Business-Validated Development — Full-stack course project, 2026

Designed and built a "Linear Gatekeeper" web app that walks founders through a fixed 23-question intake across four gated phases and auto-generates a three-document Business Suite files (Business Plan, MRD, PRD), preventing the hallucination and inconsistent-output failures common to unguided LLM business-plan generation.

- Architected a trust boundary separating deterministic computation from LLM narration: a pure TypeScript layer computes market sizing (TAM to SAM to SOM), a month-by-month revenue projection, and 12 financial formulas before the model sees any figure, with confidence-tier propagation preventing invented or corrupted numbers from reaching a document.
- Built a database-enforced citation guardrail — a Postgres CHECK constraint allow-listing 12 external data sources (World Bank, OpenStreetMap, Wikidata, etc.) — so the LLM can only cite benchmark data that actually resolved, with unvalidated/proxy/conflict labeling replacing silent gaps or fabricated figures.
- Implemented the full stack solo: Node.js/Express/TypeScript, raw PostgreSQL (no ORM, by design), React frontend, and a from-scratch PNG chart renderer to keep the local-only install dependency-free.

<https://github.com/arielzin33-creator/biz2code-v3>

HALIS — Physical AI Reception Platform for Commercial & Residential Buildings — Full-stack course project, 2026

Led system architecture and documentation for a physical AI reception platform (conversational AI + automated hardware for visitor management, parcel lockers, and access control), built as a course project through the pre-MVP design stage.

- Designed the platform's hybrid edge-cloud architecture — four-layer decomposition (interaction, orchestration engine, cloud services, edge runtime), core workflows (check-in, parcel/locker routing, access provisioning), and tech stack (React/Vite, Python/FastAPI, PostgreSQL/TimescaleDB, MQTT).
- Steered AI-assisted development of the Software Architecture Document, PRD, and UI/UX design system.
- Produced supporting business and compliance analysis (market sizing, competitive analysis, DPIA/GDPR assessment) to bring the project to pilot-readiness.

Under ongoing work

TrueTap — Browser (Chrome) Extension Against Deceptive Overlay Advertising — Personal project, 2026

A Google Chrome extension that detects and neutralizes invisible click-intercepting overlays — transparent HTML elements injected over legitimate page content (typically on streaming portals, file-hosting sites, and ad-heavy blogs) to hijack clicks and trigger pop-ups, forced redirects, or unwanted downloads.

- Defined the product mechanism and technical scope: detection and neutralization of transparent, click-capturing overlay elements distinct from classical iframe-based clickjacking.
- Tech stack - Vanilla JS Chrome extension (Manifest V3), no framework, no build tooling, no dependencies, no backend — everything is client-side (chrome.storage.local, chrome.alarms, MutationObserver).
- Directed the AI-assisted development of the proof-of-concept codebase, supervising the build personally as sole founder.

<https://github.com/arielzin33-creator/truetap>

LANGUAGES

English — Fluent | Hebrew — Native